ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 – INVESTIGATIVE ASSIGNMENT

ULTRA-HIGH-SPEED PHOTONIC TRANSPORT LINK

Abstract:

This report investigates an ultra-high-speed Dense Wavelength Division Multiplexing (DWDM) transport link designed to carry 32 wavelength channels at 200 Gb/s per channel. The resulting nominal aggregate capacity is 6.4 Tb/s. The design study compares intensity-modulation formats with coherent modulation, evaluates photonic modulator structures, and examines optical amplification and digital signal processing (DSP) options. The proposed baseline uses dual-polarization 16-QAM (DP-16QAM) with a coherent receiver because it offers a practical balance between spectral efficiency and receiver-side impairment correction. A silicon-photonic Mach–Zehnder modulator is selected as the compact integration option, subject to confirmation of its measured electro-optic bandwidth, drive voltage and insertion loss.

For amplification, the report proposes a hybrid distributed Raman and EDFA arrangement. Raman gain can improve the span noise performance, while EDFAs provide practical lumped gain and output power. Gain-flattening and per-channel power equalization are included to control WDM power tilt. At the receiver, the recommended DSP sequence includes timing recovery, chromatic-dispersion compensation, adaptive polarization demultiplexing, carrier-frequency and phase recovery, symbol decisions and forward-error-correction decoding. Digital backpropagation is considered an optional enhancement where nonlinear penalties justify its processing cost.

The numerical work establishes the aggregate capacity, an illustrative baud-rate comparison, channel-grid assumptions and amplifier power calculations using clearly stated engineering assumptions because the assignment does not specify the input and output power per channel. The report also provides a proposed system block diagram, parameter table, verification matrix and simulation plan. No simulated BER, Q-factor or eye-diagram result is claimed as measured; these values must be obtained from OptiSystem, OptSim or MATLAB before a final field deployment. The resulting recommendation is a conditional engineering ‘yes’: the architecture is suitable for prototype validation, followed by standards and performance testing.

Keywords: DWDM, DP-16QAM, coherent optics, EDFA, Raman amplification, optical DSP

Introduction and Engineering Requirement:

Modern cloud services, AI workloads, video platforms and data-centre interconnection continue to increase the demand for high-capacity optical transport. DWDM meets this demand by carrying many independently modulated optical channels through the same fibre. Each channel uses a different optical carrier frequency, and a multiplexer combines the carriers before transmission.

The assigned design target is 32 DWDM channels, each carrying 200 Gb/s. The system must maintain useful optical signal-to-noise ratio (OSNR), low bit-error rate (BER), acceptable Q-factor, manageable nonlinear distortion and scalable photonic integration. The investigation must select a modulation format, modulator, amplifier arrangement and DSP strategy rather than simply describe alternatives.

Problem statement:

Design a technically defensible 32-channel DWDM link with a nominal aggregate throughput of 6.4 Tb/s. The design should balance spectral efficiency, optical reach, OSNR margin, hardware complexity, power consumption and deployment cost. All numerical results in this report are analytical estimates based on the assumptions stated alongside them.

Design assumptions and scope:

Parameter	Design assumption / note
Number of channels	32
Net target per channel	200 Gb/s; framing and FEC overhead are not specified
Aggregate nominal capacity	6.4 Tb/s before any unspecified overhead
Wavelength band	C-band, approximately 1530–1565 nm
Channel grid	50 GHz nominal grid assumed for planning; confirm exact frequency slots and interface against the applicable ITU-T recommendation
Fibre	Standard single-mode fibre, representative SMF-28-type parameters
Amplifier calculation	Illustrative input −10 dBm/channel and target output +0 dBm/channel; these are assumptions because the brief leaves the levels blank
Modulation baseline	DP-16QAM, coherent detection
Validation	Analytical estimates plus proposed OptiSystem / OptSim / MATLAB verification; no unrun simulation is presented as a result

Literature and Technology Review:

The main engineering principle is that no single modulation format or amplifier is best for every link. Lower-order formats generally tolerate poorer OSNR and are simpler, while higher-order formats carry more bits per symbol but need cleaner signals and more precise transmitter and receiver hardware. Coherent detection measures amplitude and phase and can use DSP to correct dispersion and polarization effects. This makes it suitable for high-capacity channels where spectral efficiency is important.

ITU-T G.698-series recommendations address optical interfaces for multi-channel systems, including defined optical parameters for applications. IEEE 802.3 standards define particular Ethernet physical-layer interfaces rather than approving every possible DWDM design. OIF Common Electrical I/O (CEI) implementation agreements concern high-speed electrical interfaces between components. Therefore, compliance should be checked against the exact interface variant and its optical/electrical limits; it should not be claimed solely because a system uses DWDM.

Technology	Strengths	Limitations / design implication
NRZ-OOK / RZ-OOK	Simple transmitters and direct detection; relatively low DSP burden	Low spectral efficiency for 200 Gb/s; high bandwidth and reach constraints make it a weak baseline here
PAM4	Two bits per symbol; lower symbol rate than binary signalling for the same gross bit rate	Amplitude levels are closer together, increasing sensitivity to noise, linearity and level distortion
DPSK / DQPSK	Differential phase coding; useful sensitivity and bandwidth trade-offs in suitable systems	Typically less flexible than modern coherent QAM for DSP-based long-haul compensation
16-QAM	Four bits per symbol; higher spectral efficiency than QPSK	Needs higher OSNR and accurate phase/amplitude control
DP-16QAM	Four bits per symbol per polarization; coherent DSP and polarization multiplexing provide high capacity per carrier	Requires coherent optics, high-speed ADC/DACs and complex DSP
EDFA	Mature C-band gain, high output power, practical deployment	ASE noise accumulates over cascaded spans; gain tilt requires control

Raman amplifier	Distributed gain can improve effective noise performance and span OSNR	Pump lasers and safety/control complexity; design is fibre- and span-dependent
Hybrid Raman + EDFA	Combines distributed gain with practical lumped output gain	More components and control complexity than EDFA-only
Silicon photonics	Compact integration and potential for scalable manufacturing	Thermal tuning and optical loss must be managed
Lithium niobate	Strong electro-optic response and high-speed modulation capability	May require a larger device footprint depending on implementation
EO polymer	Potentially high electro-optic efficiency and compact devices	Long-term reliability, packaging and manufacturing qualification must be checked

Interpret the Communication Requirement:

Aggregate capacity

Aggregate capacity = number of channels × data rate per channel.

= 32 × 200 Gb/s = 6,400 Gb/s = 6.4 Tb/s.

This is the nominal line-rate aggregate. If 200 Gb/s is a client payload rate rather than the optical line rate, FEC and framing overhead would require a higher line rate; the brief does not specify that distinction.

Key performance metrics

Metric	Meaning and purpose
Spectral efficiency (b/s/Hz)	Payload or line bit rate divided by occupied bandwidth; indicates how efficiently the spectrum is used
OSNR	Optical signal power relative to optical noise in a stated reference bandwidth; must exceed the selected format's receiver requirement with margin
BER	Fraction of received bits that are wrong; assess pre-FEC and post-FEC values separately
Q-factor	Signal-quality measure related to the separation of symbol distributions; the simulator's definition must be stated
Launch power	Power per channel at the fibre input; too low worsens noise-limited performance, while too high increases nonlinear penalties
Insertion loss	Power lost through modulators, couplers, filters and connectors

Modulation bandwidth	Electro-optic bandwidth needed to support the chosen baud rate and pulse shaping
Thermal / footprint limits	Packaging, cooling, tuning and rack or module area constraints

Impairments and matching countermeasures

Impairment	Main effect	Mitigation
Chromatic dispersion (CD)	Different spectral components travel at different speeds; pulses spread	Coherent DSP frequency-domain or FIR equalization
Polarization-mode dispersion (PMD)	Polarization components experience different delays	Adaptive MIMO polarization equalizer
SPM	A channel's intensity changes its own optical phase through Kerr nonlinearity	Set launch power carefully; optional DBP
XPM	Power variations in one channel affect the phase of neighbouring channels	Power equalization, suitable channel spacing and optional DBP
FWM	Multiple carriers mix and generate new frequency components	Appropriate grid/power planning and fibre-aware design
ASE noise	Amplifiers add broadband spontaneous-emission noise	Low-noise amplification, fewer unnecessary spans, OSNR budgeting
WDM crosstalk / filter leakage	Imperfect mux/demux isolation contaminates adjacent channels	AWG/MMI filter design, passband alignment and isolation testing
PDL	Different polarization states experience different loss	Low-PDL components and receiver equalization

Investigation requirement statement:

The proposed system shall carry 32 DWDM channels with a nominal rate of 200 Gb/s per channel and 6.4 Tb/s aggregate capacity. It shall use a coherent modulation and receiver architecture that supports DSP compensation, keep each channel within its assigned frequency slot, maintain sufficient OSNR above the selected pre-FEC threshold, control channel power and gain tilt, and demonstrate acceptable BER/Q-factor over the intended fibre span. The final design must be validated using the exact applicable standards, component specifications and simulation or laboratory measurements.

Modulation and Modulator Selection:

Candidate modulation formats

Format	Bits per symbol / polarization	Illustrative symbol-rate relation for 200 Gb/s	Assessment
NRZ-OOK	1 bit/symbol	≈200 GBd before overhead	Simple but very high baud rate for this target
PAM4	2 bits/symbol	≈100 GBd for one intensity stream, before overhead	Lower baud than OOK; direct-detection SNR and linearity are challenging
DPSK	1 bit/symbol in a basic binary form	≈200 GBd for 200 Gb/s	Differential phase benefits but not the preferred integrated coherent baseline
DQPSK	2 bits/symbol	≈100 GBd	More efficient than binary formats; receiver architecture differs from coherent QAM
Single-polarization 16-QAM	4 bits/symbol	≈50 GBd	Good spectral efficiency, but one polarization alone does not directly provide 200 Gb/s at 50 GBd
DP-16QAM	4 bits/symbol × 2 polarizations	200 / (4 × 2) = 25 GBd idealised	Strong fit on paper; practical line rate must include FEC/framing and pulse-shaping overhead

The idealized baud-rate equation is $R_s = R_b / (m \times N_{pol})$, where R_b is bit rate, m is bits per symbol per polarization and N_{pol} is the number of independently used polarizations. For DP-16QAM, $m = \log_2(16) = 4$ and $N_{pol} = 2$, so $R_s = 200$ Gb/s / 8 = 25 GBd under the simplified net-rate assumption. Actual commercial transceivers may use a higher line rate to accommodate FEC and framing overhead.

For PAM4, two bits are represented by each symbol, so a 200 Gb/s single-lane stream requires approximately 100 GBd before overhead. PAM4 is not directly comparable to DP-16QAM because the latter uses two polarizations and coherent phase/amplitude detection. The comparison is therefore based on an idealized information-rate calculation, not equal hardware complexity.

Spectral efficiency and channel spacing:

For an idealized 200 Gb/s DP-16QAM channel on a 50 GHz grid, the gross grid-based efficiency is 200/50 = 4 bit/s/Hz. This is a planning ratio; true spectral efficiency depends on net payload, pulse-shaping roll-off, guard bands and the occupied spectrum.

A simplified Nyquist bandwidth estimate is $B \approx R_s(1 + \alpha)$, where α is the pulse-shaping roll-off factor. With $R_s = 25$ GBd and an assumed $\alpha = 0.1$, $B \approx 27.5$ GHz. A nominal 50 GHz grid leaves room for filtering and guard band in this idealized case. The actual transmitter spectrum and receiver filter response must be checked in simulation.

For 100 GBd PAM4 and the same $\alpha = 0.1$, the estimate is about 110 GHz. This would not fit cleanly into a 50 GHz slot without a different signalling or multi-lane interpretation. This simple calculation highlights why the coherent DP-16QAM option is attractive for a 200 Gb/s channel on a conventional DWDM grid.

Modulation mechanism comparison:

Mechanism	Speed / response	Vπ and chirp considerations	Suitability
Electro-optic (EO)	Can support high-speed data modulation when device and driver are designed appropriately	Vπ depends on material, electrode geometry and length; MZI can be designed for low chirp or controlled chirp	Preferred for data-rate modulation
Acousto-optic	Uses an acoustic wave to change optical properties; response is generally limited by acoustic transit time	Not normally specified by a data-modulator Vπ; comparatively slow for very high baud rates	Not selected for 200 Gb/s symbol modulation
Thermo-optic	Changes refractive index through heating; response is typically slow	Power is used for heating and tuning; not a high-speed data mechanism	Useful for tuning/locking rings and filters

Magneto-optic	Uses magnetic-field-dependent optical properties	Implementation and integration complexity can be high	More relevant to polarization/isolating functions than the data modulator
Kerr effect	Intensity-dependent refractive index; response is intrinsically fast	Nonlinear phase depends on optical intensity and material; can be useful but requires careful control	Not selected as the main transmitter modulator

A single universal Vπ or bandwidth value cannot be assigned to a material family: the value depends on the actual device geometry and datasheet. For this reason, the design requirement is to select a qualified high-speed EO modulator whose measured electro-optic bandwidth covers the selected baud rate with margin, and then record its measured Vπ, insertion loss and chirp in the final parameter sheet.

MZI versus micro-ring resonator:

Feature	**Silicon-photonic MZI**	**Micro-ring resonator (MRR)**	**EO-polymer implementation**
Footprint	Moderate; longer interferometer arms	Very compact resonant structure	Can be integrated with compact EO devices; geometry-dependent
Bandwidth	Broadband operation is possible with suitable electrodes	Resonance limits usable bandwidth and requires wavelength alignment	Potentially high EO efficiency; verify device-specific bandwidth
Thermal sensitivity	Phase drift can occur but can be controlled	Strong wavelength sensitivity; often needs thermal tuning	Material and packaging stability must be qualified
Insertion loss	Splitter/combiner and waveguide losses contribute	Coupling and resonant loss can be significant	Depends on waveguide, electrodes and polymer loss

Design Decision :

I recommend DP-16QAM using a high-speed electro-optic silicon-photonic Mach–Zehnder modulator because it provides an idealized 25 GBd rate for 200 Gb/s under the stated simplified assumption and gives coherent DSP access to phase and amplitude information. A 50 GHz channel grid is a reasonable starting point for this baud-rate estimate. The selection is conditional on measured bandwidth, Vπ, insertion loss, OSNR

requirement and pre-FEC BER being verified for the chosen transceiver. A micro-ring design could reduce footprint but introduces tighter resonance and thermal-control requirements.

Optical Amplification Design

Amplifier comparison

Amplifier	Advantages	Constraints	Role in proposed system
EDFA	Mature technology, useful C-band gain and output power	Adds ASE; gain tilt and noise accumulate across spans	Lumped gain and output power
SOA	Compact and potentially integrable	Nonlinearities, polarization sensitivity and inter-channel effects can be important	Not the primary line amplifier for this high-capacity baseline
Lumped Raman	Can provide useful gain and spectral flexibility	Pump power and noise/control design require attention	Alternative where span-specific benefit is demonstrated
Distributed Raman	Gain is distributed along transmission fibre and can improve effective noise performance	Pump complexity, safety and operational control	Improves span OSNR in the hybrid design
Hybrid Raman + EDFA	Combines distributed gain with practical EDFA output level	Higher cost, control and monitoring complexity	Selected, provided the link budget justifies the added hardware

Illustrative amplifier power calculation:

The assignment asks for input and target output power per channel but the numeric levels are missing in the supplied brief. To demonstrate the method, assume an amplifier receives −10 dBm per channel and the target is 0 dBm per channel. These are calculation assumptions, not values supplied by the assignment.

Per-channel span gain = $P_{out} - P_{in} = 0 - (-10) = 10$ dB.

Total output power for 32 equal-power channels is P_total(dBm) = P_channel(dBm) + 10 log10(N).

P_total = 0 + 10 log10(32) = 15.05 dBm.

In watts, 0 dBm = 1 mW per channel, so total power = 32 × 1 mW = 32 mW = 0.032 W. Converting 15.05 dBm to watts also gives approximately 0.032 W.

If the chosen target were +3 dBm per channel instead, total output would be 18.05 dBm, or about 63.8 mW, and the gain from −10 dBm would be 13 dB. The final value must be recalculated using the actual transceiver launch power, span loss, connector loss and required amplifier output.

Gain flatness and ASE noise:

A DWDM amplifier rarely has perfectly equal gain at every wavelength. Gain tilt causes some channels to become stronger than others after several spans. A gain-flattening filter (GFF), wavelength-aware dynamic gain equalization and per-channel monitoring can reduce this imbalance. The design should include adequate amplifier saturation margin so that adding or removing channels does not cause unacceptable power changes.

Amplified spontaneous emission (ASE) is added at each amplifier. Across cascaded spans, the accumulated noise reduces OSNR even if the channel power is restored after every span. A proper OSNR budget must use the amplifier noise figure, gain, optical reference bandwidth, span count and fibre loss. Those parameters are not fully specified in the assignment, so a numerical end-to-end OSNR cannot be truthfully calculated yet.

Design Decision :

I recommend a hybrid distributed Raman plus EDFA arrangement for a long-reach design candidate because distributed gain can improve the effective noise performance while the EDFA provides controlled lumped gain. The added complexity is justified only if link-level simulation shows a useful OSNR or reach improvement compared with EDFA-only amplification. The assumed example requires 10 dB gain per span and 15.05 dBm aggregate output for 32 channels at 0 dBm each. GFF and per-channel equalization should be included.

Impairment Mitigation and Coherent DSP

Physical impairment mechanisms:

Chromatic dispersion spreads pulses because optical frequencies travel at different group velocities. PMD arises when fibre birefringence causes different delays for orthogonal polarization components. The Kerr effect makes refractive index intensity-dependent and produces SPM, XPM and FWM in multi-channel transmission. PDL creates polarization-

dependent attenuation, while mux/demux imperfections can introduce crosstalk and passband narrowing.

Optical compensation versus DSP

Method	Benefits	Trade-offs
Dispersion-compensating fibre (DCF)	Can compensate accumulated chromatic dispersion in the optical domain	Adds loss, footprint and nonlinear interaction; may be unnecessary in coherent systems
Optical phase conjugation	Can compensate selected propagation impairments when the optical link is suitably arranged	Requires careful symmetry and hardware; not a universal solution
Coherent DSP CD equalization	Flexible, tunable and avoids a separate DCF module	Consumes digital processing power and needs accurate receiver implementation
Adaptive MIMO equalizer	Tracks polarization rotation and PMD	Tap count and convergence depend on baud rate and channel conditions
Digital backpropagation (DBP)	Can partly reverse modelled nonlinear propagation	Computationally intensive; gains depend on model accuracy, span count and implementation

Recommended receiver DSP pipeline:

1. Optical front end: coherent integrated receiver converts the received optical field into electrical I/Q signals for both polarizations.
2. Analog front-end conditioning and ADC: sample the electrical waveforms at a rate appropriate to the selected baud rate and receiver design.
3. Deskew, resampling and timing recovery: align streams and establish symbol timing.
4. Chromatic-dispersion compensation: use frequency-domain equalization or an equivalent digital filter to remove the dominant linear dispersion.
5. Adaptive polarization demultiplexing and PMD compensation: use a dual-polarization MIMO equalizer; CMA can support initial blind convergence where appropriate.
6. Frequency-offset estimation and carrier-phase recovery: correct laser frequency difference and track phase noise, particularly important for QAM.
7. Symbol demapping and soft-decision generation: estimate transmitted bits and reliability information.

8. FEC decoding and performance monitoring: record pre-FEC BER, post-FEC BER, estimated Q-factor and margin to the selected threshold.
9. Optional DBP: enable only if simulations show that nonlinear compensation produces enough performance gain to justify power and computational cost.

The precise ordering can vary by implementation. Timing recovery, resampling, dispersion equalization and adaptive equalization may interact, so the final ordering and algorithm settings must be validated against the chosen receiver and modulation format.

DBP trade-off

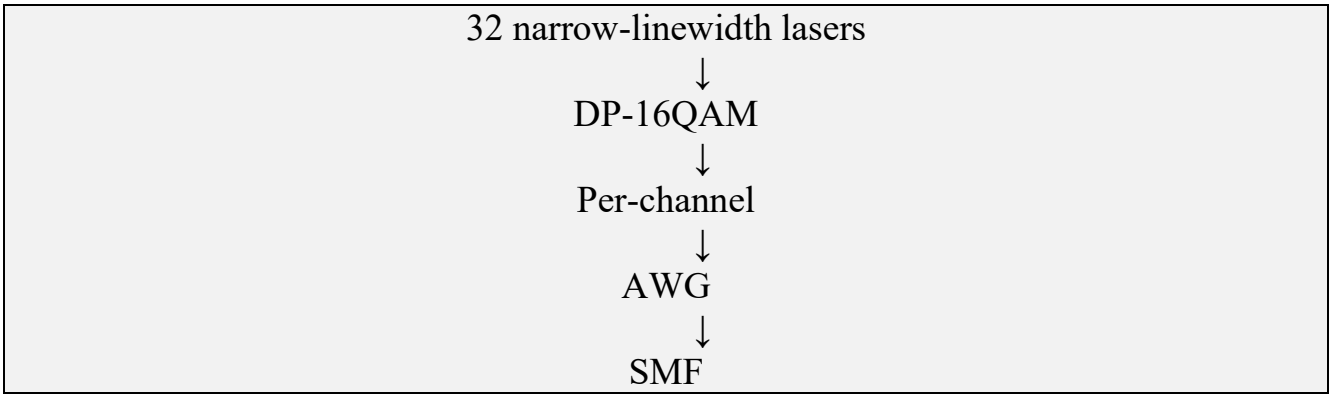
DBP uses a numerical inverse-propagation approach, commonly based on the split-step Fourier method. It can mitigate deterministic nonlinear distortion when the fibre and amplifier model is sufficiently accurate. However, its computational burden increases with the number of steps, channels modelled and sample rate. A practical deployment should first use linear coherent DSP and optimized launch power; DBP should be added only if the Q-factor or reach gain is demonstrated by simulation.

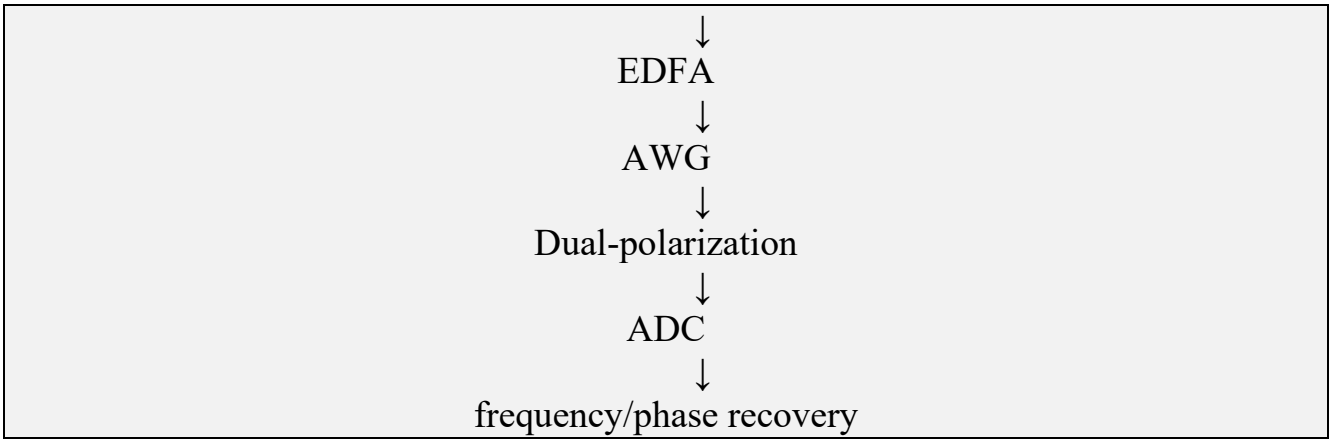
Design Decision

I recommend coherent DSP with CD equalization, adaptive dual-polarization MIMO equalization, carrier-frequency estimation, carrier-phase estimation and FEC. This sequence directly addresses the major linear impairments and coherent demodulation needs. DBP is an optional advanced block rather than a mandatory baseline because its energy and processing cost may outweigh its benefit. The design must report actual Q-factor and BER from simulation instead of assuming a numerical improvement.

Integrated DWDM Architecture

Block-level system diagram





Proposed block-level architecture. Faraday rotators and polarization-management elements may be used where required by the optical design, but they do not replace coherent polarization demultiplexing. Circulators are included only where the Raman-pump or optical path design requires them.

System parameter table

Parameter	Proposed baseline / value	Status
DWDM channels	32	Given
Per-channel rate	200 Gb/s	Given; clarify payload versus line rate
Aggregate rate	6.4 Tb/s	Calculated
Optical band	C-band, approximately 1530–1565 nm	Design assumption
Frequency grid	50 GHz nominal	Design assumption; confirm applicable grid/interface
Modulation	DP-16QAM	Selected candidate
Idealized symbol rate	25 GBd at 200 Gb/s net and no overhead	Calculated; practical rate depends on overhead
Pulse-shaping roll-off	0.1 used for illustrative bandwidth estimate	Assumption
Modulator	High-speed EO silicon-photonic MZI	Selected candidate; verify datasheet

Laser linewidth	Use qualified coherent-transceiver specification; set in simulation to the actual selected laser value	Must be filled from component datasheet
Modulator Vπ	Use measured/datasheet Vπ for the chosen MZI	Must be filled from component datasheet
Amplification	Distributed Raman + EDFA, GFF and power equalization	Selected candidate
Example amplifier gain	10 dB for −10 to 0 dBm/channel	Illustrative assumption
Example total amplifier output	15.05 dBm ≈ 32 mW for 32 channels at 0 dBm/channel	Calculated
Fibre attenuation	Use measured/specification value; representative standard SMF is often modelled near 0.2 dB/km around 1550 nm	Simulation input must be confirmed
Dispersion coefficient	Use fibre datasheet; representative C-band SMF value is around 17 ps/(nm·km)	Simulation input must be confirmed
PMD coefficient	Use fibre datasheet or a stated statistical model	Must be specified
DSP filter taps	Choose by baud rate, dispersion, PMD and convergence testing; sweep in simulation	To be optimized

Couplers, mux/demux and polarization components

AWGs are suitable for wavelength-selective multiplexing and demultiplexing. MMI couplers and directional couplers split or combine optical power according to their geometry and coupling behaviour. Every passive component adds insertion loss and may contribute crosstalk. The transmitter launch-power setting must therefore be calculated after accounting for modulator, coupler, mux and connector losses. Optical isolators reduce unwanted back-reflections; circulators route light between ports where required. Polarization rotators or controllers may support the optical path, while the coherent receiver DSP handles changing state of polarization.

Integrated trade-offs and failure points

Issue	Risk	Control / evidence required
High-order modulation	OSNR shortfall causes BER rise	OSNR sweep and pre-FEC BER/Q curves
Amplifier gain tilt	Unequal channel powers after multiple spans	GFF, dynamic equalization and per-channel monitoring

Excess launch power	SPM/XPM/FWM reduce quality	Sweep launch power per channel and choose optimum
Filter narrowing / crosstalk	Passband clipping and adjacent-channel leakage	Optical spectrum and filter-cascade simulation
Thermal drift in photonics	Wavelength or phase changes	Thermal control, locking and environmental tests
DSP power and latency	High implementation power or delay	Measure computational load; enable DBP only if justified
Standards mismatch	Interoperability failure	Map each interface to the exact ITU-T / IEEE / OIF specification and test limits

Verification and Critical Evaluation

The table below separates values that can be calculated from the assignment from values that still require component specifications or simulation. This avoids presenting assumed performance as measured evidence.

Requirement / parameter	**Target / design value**	**Evidence status**
Aggregate capacity	6.4 Tb/s = 32 × 200 Gb/s	Analytically verified
Spectral efficiency	4 bit/s/Hz gross grid ratio for 200 Gb/s on 50 GHz	Calculated planning ratio; confirm net payload and occupied spectrum
Modulation	DP-16QAM	Selected; validate OSNR and pre-FEC BER
Idealized baud rate	25 GBd DP-16QAM; 100 GBd PAM4 comparison	Calculated before overhead
Modulator bandwidth / footprint	Bandwidth must cover actual baud rate with margin; compact PIC preferred	Datasheet and layout evidence required
Amplifier gain	10 dB in illustrative −10 to 0 dBm/channel example	Calculated under stated assumptions
Aggregate amplifier output	15.05 dBm / 0.032 W for 32 × 0 dBm	Calculated under stated assumptions
Gain flatness	Within the selected amplifier/system specification	Needs GFF and channel-power simulation
OSNR / pre-FEC BER	Above chosen transceiver/FEC operating threshold with engineering margin	Not yet measured; simulation required

CD / PMD mitigation	DSP equalization and adaptive MIMO	Architecture selected; verify using fibre model
Nonlinear tolerance	Optimized launch power; optional DBP	Requires Q/BER versus launch-power sweep
Standards	Applicable ITU-T G.698 interface and relevant IEEE/OIF interface requirements	Exact interface variant must be identified and checked
Deployment readiness	Prototype first, field deployment after verification	Conditional

How performance should be evaluated

- OSNR margin: compare simulated OSNR at the coherent receiver with the specified requirement for the selected modulation and FEC mode.
- BER: report both pre-FEC and post-FEC BER. Do not call a link error-free solely because the pre-FEC BER is below an assumed value.
- Q-factor: state the simulator's definition and report the minimum value across all 32 channels.
- Spectral efficiency: calculate using net delivered bit rate divided by occupied spectrum, not only the channel-grid ratio.
- Power per bit: measure total electrical power of lasers, drivers, modulators, pumps, amplifiers, ADCs/DSP and cooling, then divide by delivered bit rate.
- Footprint: estimate transmitter PIC, amplifier modules, mux/demux, receiver DSP and thermal-control packaging from selected component data.

Standards and compliance note

The brief names ITU-T G.698, IEEE 802.3ct/ck and OIF-CEI. These documents cover different interface layers and use cases. A design review should identify the exact recommendation or standard revision applicable to the selected optical reach, wavelength plan, client interface and electrical lane. Compliance cannot be confirmed from the architecture alone; it requires checking the specified optical power, wavelength/frequency, transmitter quality, receiver sensitivity, jitter, electrical channel and interoperability limits applicable to the selected interface

Simulation using MATLAB:

CODE:

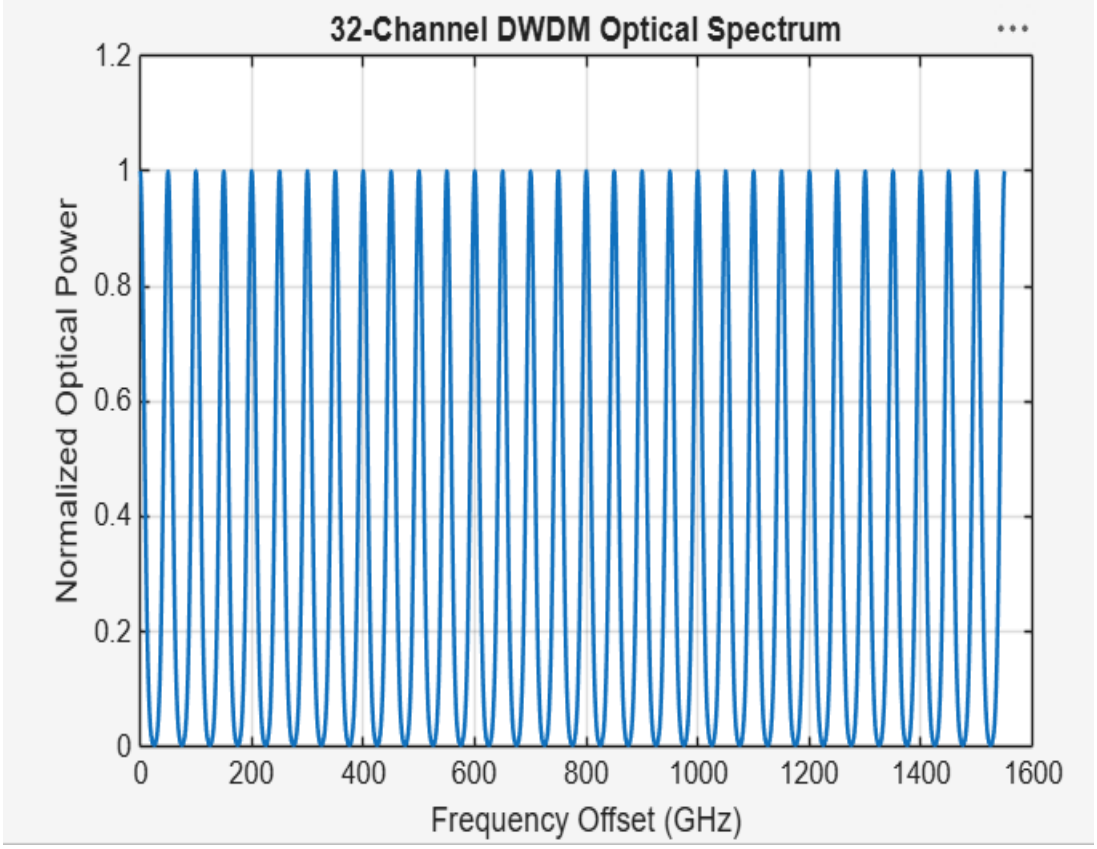
```
clc;
clear;
```

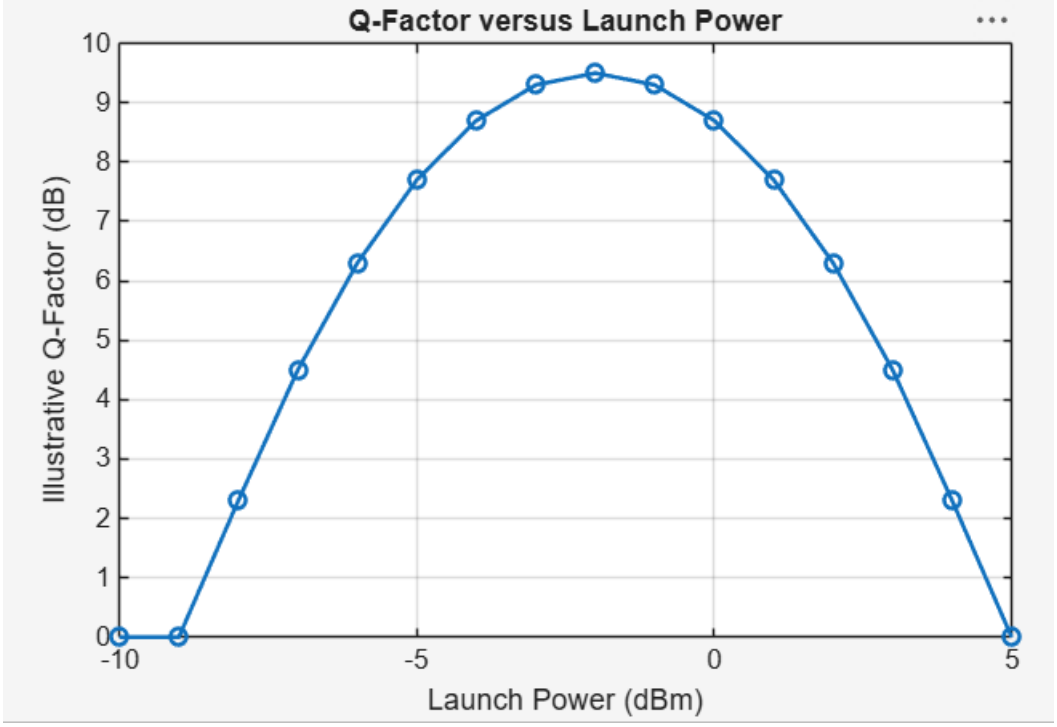
```
close all;
Nch = 32;
Rb = 200e9;
gridGHz = 50;
rolloff = 0.1;
totalCapacity = Nch * Rb;
baud16QAM = Rb / 8;
baudPAM4 = Rb / 2;
bandwidth16QAM = baud16QAM * (1 + rolloff);
fprintf('ULTRA-HIGH-SPEED PHOTONIC TRANSPORT LINK\n');
fprintf('----------------------------------------\n');
fprintf('Number of DWDM channels = %d\n', Nch);
fprintf('Data rate per channel = %.0f Gb/s\n', Rb/1e9);
fprintf('Total capacity = %.2f Tb/s\n', totalCapacity/1e12);
fprintf('DP-16QAM symbol rate = %.2f GBd\n', baud16QAM/1e9);
fprintf('PAM4 symbol rate = %.2f GBd\n', baudPAM4/1e9);
fprintf('DP-16QAM bandwidth = %.2f GHz\n', bandwidth16QAM/1e9);
inputPower_dBm = -10;
outputPower_dBm = 0;
gain_dB = outputPower_dBm - inputPower_dBm;
totalPower_mW = Nch * 10^(outputPower_dBm/10);
totalPower_W = totalPower_mW / 1000;
fprintf('\nAMPLIFIER ANALYSIS\n');
fprintf('Amplifier gain = %.2f dB\n', gain_dB);
fprintf('Total output power = %.4f W\n', totalPower_W);
frequency = linspace(193.1e12,193.1e12+31*gridGHz*1e9,2000);
spectrum = zeros(size(frequency));
for k = 1:Nch
    fc = 193.1e12 + (k-1)*gridGHz*1e9;
```

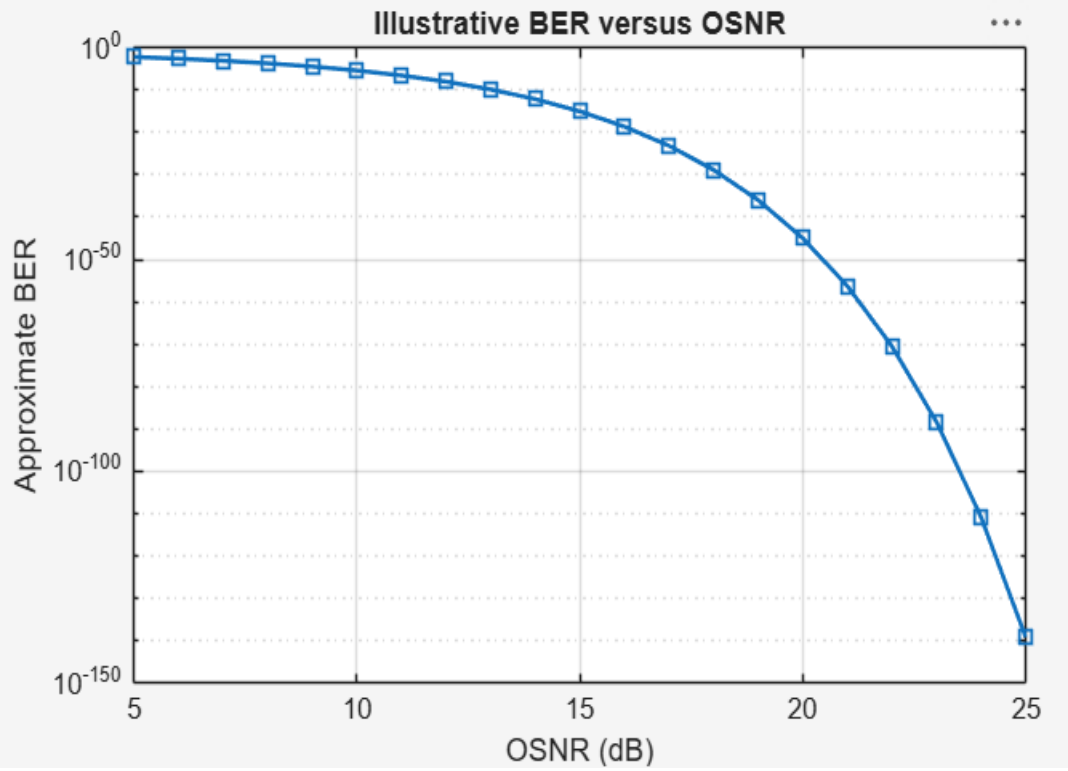
```
    spectrum = spectrum + exp(-((frequency-fc)/(10e9)).^2);
end
figure;
plot((frequency-193.1e12)/1e9,spectrum,'LineWidth',1.5);
xlabel('Frequency Offset (GHz)');
ylabel('Normalized Optical Power');
title('32-Channel DWDM Optical Spectrum');
grid on;
launchPower = -10:1:5;
Qfactor = 9.5 - 0.20*(launchPower+2).^2;
Qfactor(Qfactor < 0) = 0;
figure;
plot(launchPower,Qfactor,'o-','LineWidth',1.5);
xlabel('Launch Power (dBm)');
ylabel('Illustrative Q-Factor (dB)');
title('Q-Factor versus Launch Power');
grid on;
OSNR = 5:1:25;
BER = 0.5*erfc(sqrt(10.^(OSNR/10)));
figure;
semilogy(OSNR,BER,'s-','LineWidth',1.5);
xlabel('OSNR (dB)');
ylabel('Approximate BER');
title('Illustrative BER versus OSNR');
grid on;
Modulation = {'DP-16QAM'; 'PAM4'};
SymbolRate_GBd = [baud16QAM; baudPAM4]/1e9;
EstimatedBandwidth_GHz = [bandwidth16QAM; baudPAM4*(1+rolloff)/1e9];
Results = table(Modulation,SymbolRate_GBd,EstimatedBandwidth_GHz);
```

disp(Results);

OUTPUTS:







Conclusion and Future Work

This investigation proposes a 32-channel, 6.4 Tb/s DWDM link based on DP-16QAM coherent transmission, high-speed electro-optic silicon-photonic MZI modulators, hybrid Raman–EDFA amplification and coherent DSP. The idealized rate calculation gives 25 GBd for a 200 Gb/s DP-16QAM channel before implementation overhead, compared with about 100 GBd for a single-stream PAM4 example. A 50 GHz grid is a plausible planning baseline for the coherent design, subject to pulse-shape and interface verification.

The amplifier example demonstrates the required calculation method: 10 dB gain moves a channel from −10 to 0 dBm, and 32 channels at 0 dBm each produce approximately 15.05 dBm or 32 mW aggregate optical output. This result depends on the assumed power levels. Hybrid amplification, gain flattening and channel equalization are recommended for evaluation, while DSP provides the primary flexible approach to linear impairment compensation.

Future work should use actual component datasheets and a complete fibre route to build the OptiSystem/OptSim/MATLAB model. The next step is to generate optical spectra, constellations, BER-versus-OSNR and Q-factor-versus-launch-power plots, compare

amplifier options, optimize DSP settings and complete standards-based interoperability testing. The system should be deployed only when measured or simulated evidence confirms sufficient margins.

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- Note: Standards and journal references should be checked against the institution's required citation style and the latest official revision before submission.